

ANOVA in Regression — Line-by-Line Easy Calculation

This PDF explains exactly how to calculate the Regression, Residual, and Total rows in the ANOVA table.

1. First understand the ANOVA table

Source	df	SS	MS	F	Significance F
Regression	1	15961.40421	15961.40421	24912.38139	9.9812E-23
Residual	13	8.329121632	0.640701664		
Total	14	15969.73333			

2. What each column means

df = degrees of freedom.

SS = Sum of Squares. It measures variation.

MS = Mean Square = $SS \div df$.

F = Regression MS $\div$ Residual MS.

Significance F = the p-value associated with the F statistic.

3. Important values from your regression

Number of observations: $n = 15$

Number of independent variables: $k = 1$

Total degrees of freedom = $n - 1 = 15 - 1 = 14$

Regression degrees of freedom = $k = 1$

Residual degrees of freedom = $n - k - 1 = 15 - 1 - 1 = 13$

4. Calculate Total SS

Total SS measures the total variation in Sales around the mean Sales.

Formula: $SST = \sum(Y - \bar{Y})^2$

From the Sales data, Mean Sales = 157.4667.

After calculating $(Y - \bar{Y})^2$ for all 15 observations and adding them:

$SST = 15969.73333$

5. Calculate Residual SS

Residual SS measures the variation left unexplained by the regression line.

First calculate the predicted Sales for each observation using:

$\hat{Y} = 74.5566 + 3.01857X$

Then calculate the error for every row:

Error = $Y - \hat{Y}$

Then square every error:

Error² = $(Y - \hat{Y})^2$

Add all 15 squared errors:

$SSE = \sum(Y - \hat{Y})^2 = 8.329121632$

6. Calculate Regression SS

Regression SS is the variation explained by the regression model.

The easiest formula is:

$$SSR = SST - SSE$$

Put your values:

$$SSR = 15969.73333 - 8.329121632$$

$$SSR = 15961.40421$$

7. Check the SS relationship

A very important ANOVA rule is:

$$\text{Total SS} = \text{Regression SS} + \text{Residual SS}$$

$$15969.73333 = 15961.40421 + 8.329121632$$

$$15969.73333 \approx 15969.73333 \checkmark$$

So the ANOVA SS values are correct.

8. Calculate Regression MS

Formula:

$$\text{MS Regression} = \text{SS Regression} \div \text{df Regression}$$

$$= 15961.40421 \div 1$$

$$= 15961.40421$$

Because the regression df is 1, the Regression SS and Regression MS are the same.

9. Calculate Residual MS

Formula:

$$\text{MS Residual} = \text{SS Residual} \div \text{df Residual}$$

$$= 8.329121632 \div 13$$

$$= 0.640701664$$

10. Calculate Total MS

In Excel's ANOVA table, Total MS is normally left blank.

If you want to calculate it manually:

$$\text{Total MS} = \text{Total SS} \div \text{Total df}$$

$$= 15969.73333 \div 14$$

$$\approx 1140.69524$$

But Excel does not display Total MS in the standard regression ANOVA table.

11. Calculate F statistic

The F statistic compares explained variation with unexplained variation.

Formula:

$$F = \text{MS Regression} \div \text{MS Residual}$$

$$= 15961.40421 \div 0.640701664$$

$$= 24912.38139$$

12. Calculate Significance F

Significance F is the p-value for the overall F-test.

For this regression:

$$F = 24912.38139$$

$$\text{Numerator df} = 1$$

$$\text{Denominator df} = 13$$

Using the F-distribution right-tail probability:

$$\text{p-value} = P(F_{1,13} \geq 24912.38139)$$

$$\text{Significance F} \approx 9.9812 \times 10^{-23}$$

This is an extremely small p-value, so the overall regression is statistically significant.

13. Understand the ANOVA table in one picture

TOTAL variation in Sales

↓

Total SS = 15969.73333

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Regression SS Residual SS

15961.40421 8.329121632

Explained variation Unexplained variation

14. Final table with every calculation

Source	df	SS	MS	F	Significance F
Regression	k = 1	$15969.73333 - 8.329121632 = 15961.40421$	$15961.40421 \div 1 = 15961.40421$	$15961.40421 \div 0.640701664 = 24912.38139$	9.9812E-23
Residual	15-1-1 = 13	8.329121632	$8.329121632 \div 13 = 0.640701664$	—	—
Total	15-1 = 14	15969.73333	—	—	—

15. Easy way to remember

1. Total SS = total variation.
2. Residual SS = error/unexplained variation.
3. Regression SS = Total SS – Residual SS.
4. MS = SS ÷ df.
5. F = Regression MS ÷ Residual MS.
6. Significance F = F-test p-value.
7. df Regression = k = 1.
8. df Residual = n – k – 1 = 13.
9. df Total = n – 1 = 14.

Final answers

ANOVA item	Answer
Regression df	1
Regression SS	15961.40421
Regression MS	15961.40421
Residual df	13
Residual SS	8.329121632
Residual MS	0.640701664
F	24912.38139
Significance F	9.9812E-23
Total df	14
Total SS	15969.73333